Artem Kirienko

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Career Objective

Performance-focused software engineer with 2 years of practical C++ development experience, including optimizing large-scale, low-latency applications at Apple and building a custom DirectX rendering engine from the ground up. Seeking to apply deep expertise in 3D game math and GPU rendering to develop innovative rendering techniques and deliver cutting-edge real-time graphics technology.

Technical Summary

Programming Languages: C, C++, HLSL, C#.

Frameworks: DirectX 11, CUDA, Win32, .Net, Box2D, GLTF/GLB.

Software: Visual Studio, Xcode, Unreal Engine 5, Intel VTune.

Version Control: Perforce, Git.

Platforms: Windows, iOS, tvOS, macOS.

Core Focus Areas: Real-Time Game Engine Development, Concurrent Programming, Shading Languages, Advanced Computer Graphics/3D Rendering (Deferred Volumetric Lighting), Multithreaded Rendering, 3D Game Math, GPGPU Programming, Mobile Development (iOS).

Education

DePaul University — Chicago, IL
Master of Science in Software Engineering — Mar 2025

- Concentration in Real-Time Game Systems

University Of South Florida — Tampa, FL
Bachelor of Science in Computer Science — Aug 2022

Work Experience

Apple Inc. — San Diego, CA
AirPlay Performance Software Engineer — May 2025 – Present

- Led AirPlay performance optimization effort in a multithreaded, multiprocessing, and network environment.
- Owned the design and implementation of performance profiling and monitoring tools that identified critical bottlenecks, enabling optimizations deployed to millions of users worldwide.
- Leveraged the above performance tooling to minimize variability and ensure consistent application performance across diverse hardware configurations.
- Analyzed and optimized application memory footprint, CPU utilization, instruction count, and power efficiency across Apple's embedded platforms.
- Partnered with cross-functional teams to develop and optimize SPIs, ensuring strong and consistent performance across diverse real-world use cases and platforms during integration.
- Evaluated performance implications of upcoming features during system architecture planning with senior engineers and engineering managers.

DePaul University — Chicago, IL
Graduate Research Assistant — Sep 2024 – Mar 2025

- Contributed to a real-time multi-drone ecosystem simulation for **NASA-funded** project in Unreal Engine 5.
- Owned the C++ implementation of drone camera live video streaming feature using RTP and GStreamer.
- Ensured the video's fast network transport and smooth end user experience in an UE5 simulation.
- Distributed video encoding to multiple UE5 threads, resulting in **15% increase in framerate**.

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Relevant Development Experience

More information, videos, and UMLs of my projects can be found at archangelcode.github.io/portfolio/

Real-Time 3D Game Engine `[C++/HLSL/DirectX 11]`

- Engine Core:
  - Architected a real-time DirectX game engine from the ground up, implementing 20+ HLSL shaders (Vertex, Pixel, Compute) managing complex rendering pipelines.
  - Created a C++ framework to efficiently process and send scene geometry to the GPU.
  - Engine capable of mesh rendering, affine object transformations, texturing, perspective/orthographic cameras, frustum culling, skinning, animation, and lighting.
  - Introduced an offline tool to convert mesh/texture data from .glb files to engine-compatible format.
- Multithreaded Rendering:
  - Engineered Multithreaded Rendering pipeline by separating update() and render() into dedicated threads, and synchronized them with thread-safe message queues, **reducing frame time by ~9x**.
- Lighting:
  - Programmed Volumetric Deferred Lighting technique using light-volume meshes, **increasing scene light capacity by ~25x** from ~360 to ~10,000 before noticeable framerate loss.
- Animation:
  - Established GPU-accelerated skeletal animation using **DirectCompute**, batching multiple skeletons per dispatch to maximize GPU occupancy and parallel throughput.
  - Coded 2D blend trees to smoothly blend 2 different animations over a variable period of time.
  - **Reduced GPU memory usage by ~3x** by compressing animation clips using linear and spherical interpolation while preserving visual quality.
- 3D Math Library:
  - Developed a **SIMD-enhanced 3D math library** supporting vector, matrix and quaternion operations, for the engine achieving 4x-16x faster math performance in various use cases.
  - Specialized matrix multiplication paths by transform type, eliminating unnecessary floating-point operations and **accelerating affine transform calculations by ~2-6x**.
  - Implemented Taylor-polynomial sine and cosine approximations that outperformed C++ standard-library equivalents by 2x, **optimizing rotation-matrix construction by ~3-4x**.

Space Invaders Game Remake `[C#/.NET]`

- Developed a clone of *Space Invaders* in C#, applying **12 object-oriented software design patterns**.
- Coded gameplay systems for sprite rendering, animation, collisions, input, and game state management.
- **Reduced** runtime **memory allocations** by implementing an object pool for reusable game objects.
- Designed an object creation pipeline for aliens, shields, ships, and missiles using the Factory pattern.
- Built an optimized collision system using early-out method with the Visitor and Composite patterns.
- Applied the Flyweight pattern to efficiently render fonts, including scores, level labels, and menu options.

Angry Birds Game Remake `[C++/Box2D]`

- Developed a Windows clone of *Angry Birds* in C++ using Box2D physics engine.
- Implemented rigid-body physics, including gravity, impulses, mass, acceleration, and restitution.
- Processed Box2D collision events to trigger sound effects, apply structural damage, and update bird health.
- Constructed a reusable particle system using dynamic Box2D bodies to power the black bird's explosion and collision-driven feather effects.